RISK MANAGEMENT REPORT

TYPE	Grader
MAKE	Caterpillar
MODEL	140M Grader
ASSET NUMBER	DEMO-001
REGISTRATION	—

Report Number	RA-20260302-1428
Date	02 Mar 2026
State	NSW
Created By	Demo Assessor
Owner	Demo Company

SUMMARY

Risk Treatments

In Place: 73

Required: 0

Critical: 0

High: 0

Medium: 0

Low: 0

TABLE OF CONTENTS

SECTION 1 IMPORTANT INFORMATION

This report was generated by CivDocs on 02 Mar 2026.

All operators of this item of plant must read and understand this report prior to operating this item of plant. This report pertains to this item of plant as it appeared on the day of inspection.

The safety hazards associated with the operating and maintaining of this item of plant have been identified as far as practical by visual inspection. The condition of this item of plant will change with use. No physical testing has been conducted (e.g. wire rope tests, stress tests, structural/non-destructive tests, noise tests, vibration tests, brake tests, insulation tests etc.) unless stated otherwise in the notes.

Controls outlined in both Section 4 & 5 of this report must be maintained at all times whilst this item of plant is in operation.

Any information contained in the notes section of this report shall be read in conjunction with Section 3. Any information relating to standard features have been supplied via the manufacturer and shall be used as a guide only until verified.

Additional risk assessment may be required, specific to the operating environment, for this item of plant.

All operators and maintenance personnel must be appropriately trained in the use and maintenance of this item of plant.

MACHINE IDENTIFICATION

Make / Model	Caterpillar – 140M Grader
Type	Grader
Asset Number	DEMO-001
State	NSW
Owner	Demo Company
Assessor	Demo Assessor

NOISE

Manufacturer's specified noise level dBA	85

BLADE

Blade height (mm)	610
Blade length (mm)	3658
Blade lift (mm)	610
Blade thickness (mm)	32
Blade tilt Fwd/Back (deg)	40 deg / 5 deg

BODY TYPE

Articulated / Rigid	Articulated
Articulation, either side (deg)	20

CAPACITIES

Fuel Tank Capacity (L)	265
Hydraulic Oil Tank Capacity (L)	95

DRIVES

Drive	4WD

ENGINE

Engine Make & Model	Cat C9.3
Engine Hours	5000

PLANT CLASSIFICATION

Year	2016

TRANSMISSION

Maximum speed Fwd/Rev (km/h)	46 km/h
Speeds F/R	8 Fwd / 6 Rev
Transmission	Power Shift

EXTRAS FITTED

Air Conditioning

FOPS

ROPS – Cabin

RISK ANALYSIS

•CONSEQUENCE '

	1. INSIGNIFICANT Dealt with by in house first aid	2. MINOR Treated by medical professionals, hospital out patients	3. MODERATE Significant non permanent injury overnight hospital stay	4. MAJOR Extensive permanent injury eg. Loss of fingers, extended hospital stay	5. CATASTROPHIC Death, permanent disabling injury eg. Loss of hand, quadriplegia
A. Almost certain to occur in most circum-stances	MEDIUM 8	HIGH 16	HIGH 18	CRITICAL 23	CRITICAL 25
B. Likely to occur fre-quently	MEDIUM 7	MEDIUM 10	HIGH 17	HIGH 20	CRITICAL 24
C. Possibly and likely to occur at sometime	LOW 3	MEDIUM 9	MEDIUM 12	HIGH 19	HIGH 22
D. Unlikely to occur but could happen	LOW 2	LOW 5	MEDIUM 11	MEDIUM 14	HIGH 21
E. May occur but only in rare circumstances	LOW 1	LOW 4	LOW 6	MEDIUM 13	MEDIUM 15

RISK EVALUATION

CRITICAL	Act immediately to mitigate risk. Implement risk treatment(s) in accordance with the risk treatment table below.
HIGH	Act immediately to mitigate risk. Implement risk treatment(s) in accordance with the risk treatment table below. If the appropriate risk treatments are not immediately accessible interim risk treatment strategies. Permanent risk treatments must be implemented within one week.
MEDIUM	Take reasonable steps to mitigate and monitor the risk. Implement risk treatment(s) in accordance with the risk treatment table below. Permanent risk treatments must be implemented within one month.
LOW	Take reasonable steps to mitigate and monitor the risk. Implement risk treatment(s) in accordance with the risk treatment table below. Permanent risk treatments must be implemented within three months.

RISK TREATMENT

Selecting the most appropriate risk treatment option involves balancing the costs and efforts of implementation against the benefits derived, with regard to legal, regulatory and other requirements. (source AS/NZS ISO 31000:2018)

Eliminate	Eliminate the risk source.
Substitute	Provide an alternative that is capable of performing the same task which is safer.
Isolate	Isolate people from the hazard.
Engineering	Provide or construct a physical barrier or guard.
Administration	Develop policies, procedures, practices and guidelines in consultation with employees to mitigate the risk. Provide training, instruction and supervision about the risk source.
Personal protective	Provide personal protective equipment to protect the individual from the risk source.

SECTION 4 RISK TREATMENTS REQUIRED

This section pertains to hazards created by use of this item of plant which currently do not have risk treatments in place. The recommended risk treatment measures must be developed, implemented and validated as effective prior to the operation, maintenance or testing of this item of plant.

No risk treatments required — all assessed items are compliant.

SECTION 5 RISK TREATMENTS IN PLACE

This section pertains to risk treatments currently in place on this item of plant. All operators must read and understand the entire contents of this section prior to operating this item of plant. These treatments or equivalent must remain in place at all times whilst this item of plant is in operation.

HAZARD(S)	Prelim. Risk Rating	Residual Risk Rating
INCORRECT OPERATION	HIGH 22	MEDIUM 15

Risk Treatments in Place: Operation Handbook

The manufacturer's operation handbook has been supplied for this item of plant. This handbook must be available at all times to all potential operators and supervisory staff. All potential operators must read and be familiar with this handbook prior to operating. A complete risk assessment/Job Safety Analysis must be undertaken covering all operating processes and environments associated with this item of plant. SWMS should be produced for specific tasks associated with use of this item of plant.

References: Work Health & Safety Act & Regulations- , Occupational Health & Safety Act & Regulations

HAZARD(S)	Prelim. Risk Rating	Residual Risk Rating
INCORRECT OPERATION	HIGH 22	MEDIUM 15

Risk Treatments in Place: Maintenance Manual

The manufacturer's maintenance manual(s) has been supplied for this item of plant. These manual(s) must be available at all times to all users and maintenance staff of this item of plant. All users and maintenance staff must read and be familiar with these handbook(s) prior to maintaining or repairing this item of plant. A complete risk assessment/JSEA must be undertaken covering all inspection, maintenance, servicing and transportation requirements of this piece of plant prior to use.

References: Work Health & Safety Act & Regulations- , Occupational Health & Safety Act & Regulations

HAZARD(S)	Prelim. Risk Rating	Residual Risk Rating
OPERATIONAL MALFUNCTION	HIGH 21	MEDIUM 15

Risk Treatments in Place: Service Records

Service and maintenance records are available for this item of plant. These records must continue to be managed and available at all times as part of your service and maintenance programme. (This programme includes the undertaking of regular inspections of the item of plant with specific reference to all OEM prescribed, scheduled and non scheduled service and maintenance requirements).

References: Work Health & Safety Act & Regulations- , Occupational Health & Safety Act & Regulations

HAZARD(S)	Prelim. Risk Rating	Residual Risk Rating
INCORRECT OPERATION	HIGH 22	MEDIUM 15

Risk Treatments in Place: Pre-op Checklist Grader

A pre-operational checklist is available for this Grader. All operators must complete this checklist prior to operating this Grader.

References: Work Health & Safety Act & Regulations- , Occupational Health & Safety Act & Regulations

HAZARD(S)	Prelim. Risk Rating	Residual Risk Rating
INCORRECT OPERATION	HIGH 22	MEDIUM 15

Risk Treatments in Place: SOP Grader

Safe Operation Procedures are available for this Grader. The information in the Safe Operation Procedures must be followed at all times whilst operating this Grader.

References: Work Health & Safety Act & Regulations- , Occupational Health & Safety Act & Regulations

INCORRECT OPERATION

CRITICAL 24 | MEDIUM 15

Risk Treatments in Place: Operator Competency

Only persons who are qualified, trained and experienced and/or hold the relevant certification/license can operate this item of plant. If there is not a competent/licensed person available for operation of this item of plant then only persons who are supervised by a competent/licensed person can operate this item of plant.

References: Work Health & Safety Act & Regulations- , Occupational Health & Safety Act & Regulations

CRUSHING

HIGH 22 | MEDIUM 15

Risk Treatments in Place: SWMS Loading/Unloading

Ensure that all operators follow approved SWMS/SOP when loading and unloading this machine to and from a flat top truck or trailer, low loader or tilt tray.

References: Work Health & Safety Act & Regulations- , Occupational Health & Safety Act & Regulations

CRUSHING

HIGH 22 | MEDIUM 15

Risk Treatments in Place: SWMS Load Restraint

Ensure that all operators follow the approved SWMS/SOP when restraining this machine for transport.

References: Work Health & Safety Act & Regulations- , Occupational Health & Safety Act & Regulations

CRUSHING, COLLISION

CRITICAL 25 | MEDIUM 15

Risk Treatments in Place: Brakes

The brakes fitted to this item of plant must be fully functional at all times whilst this item of plant is in operation. The brakes must be regularly inspected and tested. These inspections and tests must be documented as part of your plant safety programme.

References: AS3450

CRUSHING, COLLISION

CRITICAL 24 | MEDIUM 15

Risk Treatments in Place: Park Brake

This item of plant is fitted with a fully functional park (hand) brake which meets the following requirements – 1. Is separate to the service brakes 2. Has a device which maintains the brake in the on position until intentionally disengaged. The park brake must be regularly inspected and tested. These inspections and tests must be documented as part of your plant safety programme.

References: AS3450

SLIPPING

MEDIUM 12 | LOW 6

Risk Treatments in Place: Operator Work Area Access/Egress

Safe access and egress to the cabin/work area(s) must be maintained at all times whilst this item of plant is in operation. It must be non slip, free from damage, located at a height so as to not cause undue body stresses and strains with three points of contact available to personnel at all times. All personnel must – 1. Always face the item of plant during access and egress. 2. Always maintain three points of contact during access and egress. 3. Never carry an object(s) in his/her hand(s) during access and egress. 4. Never jump off machine.

References: AS5327

FALLING, SLIPPING

MEDIUM 12 | LOW 6

Risk Treatments in Place: Access/Egress Instruction Label

An instruction label is fitted adjacent access/egress areas to advise all personnel of the following – 1. Always face the item of plant during access and egress. 2. Always maintain three points of contact during access and egress. 3. Ensure the steps are clean. 4. Never jump off machine. This label must be clear and legible at all times whilst this item of plant is in operation.

References: ISO31000

ENTRAPMENT

HIGH 21 MEDIUM 15

Risk Treatments in Place: Two Operator Exits

The operator cabin/work area on this item of plant has a minimum of two (2) possible exits. These must be functional and accessible at all times whenever the item of plant is manned, whether during operation or maintenance activities.

References: AS5327

ENTRAPMENT

HIGH 21 MEDIUM 15

Risk Treatments in Place: Emergency Exits

The emergency exits for this item of plant meet the following requirements – 1. Clearly and legibly labelled 2. Instructions for use are clear and legible and located adjacent the exit 3. Any required tools required for use are available e.g. Emergency hammers. These exits must be legibly labelled and fully functional at all times whenever the item of plant is manned, whether during operation or maintenance activities.

References: ISO31000

POOR VISIBILITY, COLLISION

HIGH 21 MEDIUM 15

Risk Treatments in Place: Windows & Screens

Ensure the cabin/work area safety glass windows and screens are kept clean and free from cracks and other damage at all times whilst this item of plant is in use.

References: AS/NZS4024.1201, ISO20474-

POOR VISIBILITY, COLLISION

HIGH 21 MEDIUM 15

Risk Treatments in Place: Windscreen Wipers

The windscreen wipers and washers fitted to this item of plant must be fully functional at all times.

References: AS/NZS4024.1201

CRUSHING, COLLISION

HIGH 22 MEDIUM 15

Risk Treatments in Place: Loose Items - Operator Work Area

All items that could cause harm to the operator in the event of a collision or rollover are securely restrained.

References: ISO31000

INCORRECT OPERATION, SLIPPING

MEDIUM 9 LOW 4

Risk Treatments in Place: Work Area Floors

All work area floors are non-slip and free from damage & debris. Floor area must remain non-slip and free from damage & debris, including rubbish, tools and other items, at all times whilst this item of plant is in use.

References: AS/NZS4024.1201, ISO20474-

POOR VISIBILITY, COLLISION

HIGH 22 MEDIUM 15

Risk Treatments in Place: Operator Mirrors

The operator rear view mirrors fitted to this item of plant must be fully functional and kept clean at all times. There must always be at least one mirror on each side to provide rear vision to the operator to avoid striking bystanders and objects.

References: AS/NZS4024.1201, ISO14401.1

NON COMPLIANCE, STRAINS

MEDIUM 9 LOW 1

Risk Treatments in Place: Operator Seat

The operator seat fitted to this item of plant must remain free from damage and tears, and be permanently and securely fitted at all times.

References: AS/NZS4024.1401 , ISO20474-

CRUSHING

HIGH 22 MEDIUM 15

Risk Treatments in Place: Seat Belt

This item of plant is fitted with an operator seat belt. This seat belt must be free from damage, and permanently and sturdily attached at all times whilst this item of plant is in operation. Operators must use this seat belt at all times during operation.

References: ISO6683

CRUSHING, FALLING

HIGH 22 MEDIUM 15

Risk Treatments in Place: Passenger Seat Label

This item of plant is fitted with a clear hazard warning label re: Operator only, No passengers. Passengers must not be carried at anytime. This label must be clear and legible at all times whilst this item of plant is in operation.

References: AS1319- , State Health & Safety Legislation & Regulation

HEAT STROKE, DEHYDRATION

MEDIUM 9 LOW 4

Risk Treatments in Place: Air Conditioning

This item of plant is fitted with an air conditioned cabin. This air conditioned cabin helps control the air quality and temperature for the operator and also provides shade from the sun. The air conditioner must be fully functional and serviceable at all times whilst this item of plant is in operation.

References: ISO31000

BURNS

MEDIUM 12 MEDIUM 12

Risk Treatments in Place: Open Cabin

Dust, exhaust fumes, chemical fumes, sunstroke and sunburn pose serious risk to the operator both short and long term. The appropriate controls for all of these hazards must always be available whilst this item of plant is in operation. If these controls e.g. hats, sunscreen, dust masks etc are not available then operation of this item of plant must cease until these are made available to all operators.

References: ISO31000

COLLISION

HIGH 22 MEDIUM 15

Risk Treatments in Place: Phone Use Label

This item of plant is fitted with an instruction label advising that mobile phones must not be used whilst operating this machine. Accordingly all operators must not use a mobile phone at any time whilst operating machine. If phone use is necessary then operator must place machine in park configuration in a safe position prior to phone use. Operators MUST adhere to this advice at all times. This label must be clear and legible at all times whilst this item of plant is in operation.

References: AS1319- , ISO31000

POOR VISIBILITY, COLLISION

HIGH 22 MEDIUM 15

Risk Treatments in Place: Rear Camera

This item of plant is fitted with a rear camera which is suitable for day and night operations. This camera and screen must be fully functional at all times whilst this item of plant is in operation.

References: AS/NZS4024.1201, ISO31000

FALLING

HIGH 22 MEDIUM 15

Risk Treatments in Place: Handrails

All operator work platforms are fitted with guardrails which meet the following requirements: 1. All guardrails are at least 1.1m high 2. All guardrails have a mid rail 3. All sides and ends have a kick plate which is at least 100mm high. These work platforms and/or access points must have guardrails present that are fully functional and serviceable at all times whilst this item of plant is in operation.

References: AS5327

INCORRECT OPERATION, OPERATIONAL MALFUNCTION

MEDIUM 14 MEDIUM 13

Risk Treatments in Place: Restricted Access Switches

This item of plant is fitted with a device to restrict operators. A code/key must only be given to those that have appropriate experience or training.

References: AS20474.1

CRUSHING, COLLISION

MEDIUM 12 LOW 6

Risk Treatments in Place: Warning Device (Horn)

This item of plant is fitted with a fully functional audible warning device such as a horn. This must be easily accessed by the operator, and easily identifiable by nearby pedestrians. All operators should ensure the warning devices are functional at the start of each shift, by completing pre-start checklists. Warning devices should operate automatically where appropriate (eg reversing).

References: ISO7731, ISO9533

STRAINS

HIGH 19 LOW 5

Risk Treatments in Place: Controls Ergonomics

All controls including all levers, buttons, pedals, switches etc, are placed near the operator work position and are easy to reach and operate during the execution of the operator's normal duties. This applies for all persons within the 95th percentile of the normal population distribution.

References: AS/NZS4024.1901

INCORRECT OPERATION

HIGH 22 MEDIUM 15

Risk Treatments in Place: Control Labels

All controls including all levers, buttons, pedals, switches etc. are clearly labelled as to their purpose and method of operation. These labels must be maintained in a clean and serviceable condition at all times.

References: AS/NZS4024.1905

INCORRECT OPERATION

HIGH 20 MEDIUM 14

Risk Treatments in Place: Intuitive Controls

The controls fitted to this item of plant are orientated so that the movement of the control is consistent with the action of the machine e.g. moving a control lever to the left results in the machine turning to the left. This design feature must be maintained at all times whilst this item of plant is in operation.

References: AS/NZS4024.1906

INCORRECT OPERATION, SLIPPING

HIGH 17 LOW 6

Risk Treatments in Place: Control Levers/Pedals/Buttons

All controls including all levers, buttons, pedals, switches etc. must be kept non-slip and free from damage at all times.

References: AS/NZS4024.1901

CRUSHING, ENTANGLEMENT, STRIKING, COLLISION

HIGH 22 MEDIUM 15

Risk Treatments in Place: Neutral Start

This item of plant has neutral start control in place. It must be fully functional and serviceable at all times whilst this item of plant is in operation.

References: AS4024.1603

CRUSHING, COLLISION

HIGH 22 MEDIUM 15

Risk Treatments in Place: Reverse Movement Alarm

This item of plant is fitted with a reverse movement awareness alarm which is automatically activated when reverse gear is selected. This alarm must be fully functional and serviceable at all times whilst this item of plant is in operation.

References: ISO7731, ISO9533

CRUSHING, ENTANGLEMENT, FIRE

HIGH 22 MEDIUM 15

Risk Treatments in Place: Emergency Stop/Shutdown Device

This item of plant is fitted with an emergency stop/shutdown device, capable of shutting the machine down, located at the normal operating position. This device must be fully functional at all times whilst this item of plant is in operation.

References: AS20474.1

CRUSHING, ENTANGLEMENT, FIRE

HIGH 22 MEDIUM 15

Risk Treatments in Place: External Emergency Stop/Shutdown Device

This item of plant is fitted with an emergency stop/shutdown device, capable of shutting the machine down located on at least one external surface of the machine and is easily accessible. This device must be fully functional at all times whilst this item of plant is in operation.

References: AS20474.1

CRUSHING

HIGH 22 MEDIUM 15

Risk Treatments in Place: Earthmoving ROPS

A Roll Over Protective Structure (ROPS) to AS 2294, ISO 3471, ISO 12117.1 or 2 or SAE J1040 is fitted to this item of plant. A permanent label stating this standard must be attached to the structure at all times. It must also carry a warning label re: wearing of seat belts at all times whilst this item of plant is in operation, and accordingly seat belts must be worn at all times during operation.

References: AS2294, ISO3471

CRUSHING

HIGH 22 MEDIUM 15

Risk Treatments in Place: ROPS Damage

The Roll Over Protective Structure (ROPS) fitted to this item of plant must remain free from damage at all times whilst this item of plant is in operation.

References: AS2294, ISO3471

CRUSHING

HIGH 22 MEDIUM 15

Risk Treatments in Place: ROPS Label

The warning label stating that the ROPS must not be damaged at any time (including cuts, drill holes and welds) must be present, clean and legible at all times.

References: ISO3471

CRUSHING HIGH 22 MEDIUM 15

Risk Treatments in Place: ROPS Seat Belt Label

This item of plant is fitted with a ROPS and has an advisory label stating that "seatbelts must be worn". This label must be present, clean and legible at all times. All operators and passengers must wear seatbelts whilst on this item of plant.

References: AS2294, ISO3471

CRUSHING HIGH 21 LOW 5

Risk Treatments in Place: FOPS General

This item of plant is fitted with a Level I Falling Objects Protective Structure (FOPS). This structure is designed to protect the operator from small falling objects (e.g. bricks, small concrete blocks, hand tools). Before operating this item of plant a task based risk assessment must be conducted to determine the level of FOPS required. Level I - withstands 1,365 joules. Level II - withstands 11,600 joules. This task risk assessment must be undertaken before each operation, in particular when the item of plant is moved to a new location.

References: AS2294, ISO3449, ISO10262

CRUSHING HIGH 21 LOW 5

Risk Treatments in Place: FOPS Level II

This item of plant is fitted with a level II Falling Objects Protective Structure (FOPS). This structure is designed to protect the operator from heavy falling objects (e.g. trees, rocks). Care should still be exercised when operating in an area with a risk of falling objects.

References: AS2294, ISO3449

HEARING LOSS HIGH 19 MEDIUM 14

Risk Treatments in Place: Hearing Protection Label - Operator

The hazard warning label(s) re: wearing of hearing protection attached to this item of plant refer to the level of noise produced. Permanent hearing damage will result if hearing protection is not worn. These labels must be present, clear and legible at all times whilst this item of plant is in operation.

References: AS3781- , AS/NZS1269

HEARING LOSS HIGH 19 MEDIUM 14

Risk Treatments in Place: Hearing Protection Label - Bystanders

The hazard warning labels re: wearing of hearing protection for bystanders attached to this item of plant refer to the level of noise produced. Permanent hearing damage will result if hearing protection is not worn. These labels must be present, clear and legible at all times whilst this item of plant is in operation.

References: AS3781- , AS/NZS1269

POOR VISIBILITY, COLLISION HIGH 22 MEDIUM 15

Risk Treatments in Place: Machine Lights

This item of plant is fitted with self contained lighting. All of these lights must be fully functional and serviceable whilst this item of plant is in operation in areas of reduced light. If any of these lights stop working the operation must cease immediately and the faulty light be repaired before operation can continue in the areas of reduced light.

References: ISO20474-

COLLISION HIGH 22 MEDIUM 15

Risk Treatments in Place: Turning, Braking & Presence Lights

This item of plant is fitted with lighting to indicate presence, turning and braking. All of these lights must be fully functional whilst this item of plant is in operation in areas of reduced light. If any of these lights stop working the operation must cease immediately and the faulty light be repaired before operation can continue.

References: AS/NZS4024.1201

COLLISION

HIGH 22 MEDIUM 15

Risk Treatments in Place: Beacon

This item of plant is fitted with a safety beacon. This beacon must meet the following criteria at all times whilst this item of plant is in operation – Is visible up to 200m in all directions (allowing for intermittent obstruction from the plant structure whilst the plant is in operation) – Is fitted in the most appropriate location on machine to maximise visibility without risking continual damage. NOTE: more than one beacon may be fitted to meet these criteria.

References: ISO20474-

CRUSHING

MEDIUM 14 MEDIUM 13

Risk Treatments in Place: Front Grader Blade Label

The front blade on this item of plant is fitted with a hazard warning label re: crush zone, keep clear. This label must be present and fully functional and serviceable at all times.

References: AS1319- , ISO20474-

ENTANGLEMENT

HIGH 22 MEDIUM 15

Risk Treatments in Place: Engine Guards

The engine fan and alternator belts, pulleys and gears are guarded. These guards must be present and fully functional and serviceable at all times whilst this item of plant is in operation.

References: AS/NZS4024.1601

ENTANGLEMENT, SHEARING, BURNS

MEDIUM 14 MEDIUM 13

Risk Treatments in Place: Engine Guard Label

The engine fan and alternator belts, pulleys and gears are guarded. These guards have clear legible hazard warning labels re do not open or remove guards while engine is running. These labels must be present, legible and easily seen at all times whilst this item of plant is in operation.

References: AS/NZS4024.1201, AS1319-

FIRE, BURNS

MEDIUM 12 LOW 6

Risk Treatments in Place: Exhaust

The engine exhaust on this item of plant is located/fitted with a guard to prevent injury to any person and control the risk of initiating a fire. Guards must be present, fully functional and serviceable at all times whilst this item of plant is in operation.

References: AS/NZS4024.1201

FALLING, SLIPPING, TRIPPING

MEDIUM 12 LOW 6

Risk Treatments in Place: Engine Bay Access

Safe access and egress to the engine bay/work area(s) must be maintained at all times whilst this item of plant is in operation. It must be non slip, free from damage, located at a height so as to not cause undue body stresses and strains with three points of contact available to personnel at all times. All personnel must – 1. Always face the item of plant during access and egress. 2. Always maintain three points of contact during access and egress. 3. Never carry an object(s) in his/her hand(s) during access and egress. 4. Never jump off machine.

References: AS5327

ELECTRIC SHOCK, BURNS

MEDIUM 12 LOW 6

Risk Treatments in Place: Battery Cover

All batteries fitted to this item of plant are constrained to prevent displacement & fitted with a permanent sturdy cover which allows for ventilation & ensures the terminals are protected. The constraint and cover must be present and fully functional and serviceable at all times whilst this item of plant is in operation.

References: AS/NZS4024.1201

NON COMPLIANCE

HIGH 22 MEDIUM 15

Risk Treatments in Place: Battery Isolator

This item of plant is fitted with a fully functional battery isolation switch that is clearly and legibly marked and lockable in the off position. The battery isolation switch must remain clearly and legibly marked and lockable at all times whilst this item of plant is in operation.

References: AS20474.1

STRIKING, BURNS

HIGH 22 MEDIUM 15

Risk Treatments in Place: Hydraulic Hoses

This item of plant has hydraulic hoses. These hoses must be inspected each day or before each use for wear and tear. If there are visible signs of wear, immediate action must be taken to control the risk arising from this wear. These inspections must be documented. Hydraulic fluid at high pressure can penetrate the skin, never use any part of your body to check for leaks. If oil penetrates the skin seek medical advice immediately. Always use a piece of cardboard or similar to check for suspected leaks. Always wear appropriate gloves when handling hydraulic hoses. Hydraulic pressure can be stored and is a hazard. Always connect and disconnect hydraulic hoses as per the manufacturer's manual.

References: AS4024, AS2671

STRIKING, BURNS

HIGH 22 MEDIUM 15

Risk Treatments in Place: Hydraulic Damage

The hydraulic hoses to this item of plant are free from damage and protected against damage arising from contact with the plant structure. Ensure that hoses are free from damage and that protection is in place at all times whilst this item of plant is in operation. Inspection of the hydraulic hoses and protection system should be conducted regularly and documented as part of your plant safety programme.

References: AS4024, ISO4413, AS2671

STRIKING, BURNS

HIGH 22 MEDIUM 15

Risk Treatments in Place: Hydraulic Hose Failure Shield

This item of plant is fitted with a sturdy, permanent shield(s) between the hydraulic hoses and any body parts of the operator to provide protection during a hose or component failure. This shield(s) must be present and fully functional at all times whilst this item of plant is in operation.

References: AS4024, ISO4413, AS2671

FIRE

HIGH 21 MEDIUM 15

Risk Treatments in Place: Fire Extinguisher

This item of plant is fitted with an approved and maintained fire extinguisher. Fire extinguisher(s) must be present and fully functional at all times. They must be readily accessible to the operator. Regular inspections must also be carried out in accordance with the manufacturer's requirements and AS 1851 – 1995.

References: AS1851

NON COMPLIANCE

MEDIUM 14 LOW 6

Risk Treatments in Place: Engine/Motor Compartment

The engine/motor compartment is fully enclosed and lockable to prevent unauthorised access. A code/key must only be given to those that have appropriate experience or training. These points of access must remain fully lockable at all times whilst this item of plant is in operation.

References: AS20474.1

CRUSHING, COLLISION

MEDIUM 14 MEDIUM 13

Risk Treatments in Place: Grader Blade Label

The grader blade on this item of plant is fitted with a hazard warning label re: crush zone, keep clear. This label must be present and fully functional and serviceable at all times.

References: AS1319- , ISO20474-

POISONING, EXPLOSION, BURNS

HIGH 22 MEDIUM 15

Risk Treatments in Place: Tank ID Label

The tank(s) on this item of plant have clear, legible label(s) identifying their contents, and if appropriate any necessary controls re: the contents. These must be present, clear and legible at all times. (this includes radiator, hydraulic, water and petrol/diesel tanks etc.)

References: Work Health & Safety Act & Regulations- , Occupational Health & Safety Act & Regulations

CRUSHING

HIGH 21 MEDIUM 15

Risk Treatments in Place: Articulated Joint Crush Label

This item of plant has clear hazard warning labels re: crush zone, keep clear, that are attached to each side of the articulated joint. These must be present, clear and legible at all times whilst this item of plant is in operation.

References: AS/NZS4024.1201, ISO20474-

CRUSHING

HIGH 22 MEDIUM 15

Risk Treatments in Place: Articulated Joint Locking Device

This item of plant is fitted with a safety locking device to the articulated joint (either a locking arm or cylinder locking devices) and clear, legible instruction labels on both sides of the articulated joint which state that either of these devices must be engaged during any maintenance to the articulated joint. These must be present, serviceable and employed at all times whilst this item of plant is in operation.

References: AS/NZS4024.1201, AS1319-

INSTABILITY, COLLISION

HIGH 22 MEDIUM 15

Risk Treatments in Place: Tyres

The tyres and wheel components must be inspected as part of a "pre start" checklist. These inspections must be documented as part of your plant safety programme.

References: ISO31000

INCORRECT OPERATION, OPERATIONAL MALFUNCTION

HIGH 22 MEDIUM 15

Risk Treatments in Place: Drawbar Capacity Label

The item of plant has a drawbar and a hazard warning label attached adjacent the drawbar/tow hitch re: maximum towing capacity, crushing, keep clear. It must be present and fully functional and serviceable at all times.

References: ISO31000

COLLISION

MEDIUM 9 LOW 5

Risk Treatments in Place: Recovery Point Label

This item of plant is fitted with a hazard warning label adjacent the recovery tow point which states "Recovery tow point – Read manufacturer's towing instructions before towing". Failure to do so could result in DEATH or SERIOUS INJURY. This label must be clear and legible at all times whilst this item of plant is in operation.

References: ISO31000

CRUSHING

MEDIUM 14 MEDIUM 13

Risk Treatments in Place: Ripper Crush Zone Label

The rippers on this item of plant are fitted with a hazard warning label re: crush zone, keep clear. This label must be present and fully functional and serviceable at all times.

References: AS/NZS4024.1201, ISO20474-

INCORRECT OPERATION

HIGH 22 MEDIUM 15

Risk Treatments in Place: Roller Attachment

This item of plant is fitted with a roller attachment to the rear ripper bar (e.g. a "Freeroll"). Accordingly this item of plant meets the following criteria – 1. Roller does not exceed machine and ripper manufacturers weight requirements 2. Has a device in the operator cabin to select either RIPPER or ROLLER mode which is clearly labelled 3. When in Roller mode – (a) Has a visual system within clear view of the operator which provides verification that roller mode is selected (b) Has a device which limits the speed of the machine to 20 kmh/hr and (c) Has a device which releases hydraulic pressure in the ripper bar lift/lower cylinders when the service brakes are applied. All operators must be familiar with these devices prior to operation. These safety devices must be in place at all times whilst this item of plant is in operation.

References: ISO31000

CRUSHING, STRIKING, COLLISION

HIGH 22 MEDIUM 15

Risk Treatments in Place: Vehicle Frequently Reversing

The rear of this item of plant has a hazard warning label re: vehicle frequently reversing. It must be present and fully functional and serviceable at all times.

References: ISO20474-

OPERATIONAL MALFUNCTION

HIGH 22 LOW 2

Risk Treatments in Place: Major Fluid Leaks

This item of plant must remain free from leaks at all times whilst in operation (this includes engine, transmission, cooling system, air, fuel, drive line, wheel hubs, steering and hydraulics). Development of a major leak will require this item of plant to be stood-down until repaired. Minor leaks detected must be repaired within 1-14 days.

References: ISO31000

OPERATIONAL MALFUNCTION

HIGH 22 LOW 2

Risk Treatments in Place: Plant Modification

The plant is in original condition.

References: ISO31000

CURRENT OR PREVIOUS STRUCTURAL DAMAGE

CRITICAL 25 MEDIUM 15

Risk Treatments in Place: Structural Integrity

Regular checks for structural damage must be undertaken. Look for cracks in frames/chassis (current or repaired), bends or damage to structural components, etc.

References: ISO31000

SECTION 6 IMAGES AND NOTES

– No notes or images provided –

OPERATOR ACKNOWLEDGEMENT

TYPE	Grader
MAKE	Caterpillar
MODEL	140M Grader
ASSET NUMBER	DEMO-001
REPORT NUMBER	RA-20260302-1428
DATE	02 Mar 2026
OWNER	Demo Company

I the undersigned acknowledge that I have read and understand the risk management report described above.
I also acknowledge that I have received a copy of this risk management report.

Date	Name	Company / Position	Signature